

Final Report-Analyzing the Performance of Logical Consistency Tracking (or Sequential Logic) among Conversational AI

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Abstract

When interacting with conversational AI systems such as ChatGPT, users often revise prior queries, requiring models to adjust their reasoning accordingly. A robust model should recognize these changes, update its logical chain, and generate contextually consistent responses — a capability known as logical consistency tracking. This project aims to develop a dedicated dataset for evaluating AI models’ ability to adapt to dynamic user revisions. We construct dialogues with both correct and incorrect logical transitions by systematically modifying existing resources such as SNTL and DialogStudio. Using tailored evaluation metrics, we assess the performance of models including ChatGPT and LLaMA2. Finally, we apply the dataset for fine-tuning models and evaluate whether their logical consistency tracking improves post-adaptation.

1 Introduction

In recent years, logic tracking and dialogue state tracking (DST) have been extensively studied in the field of conversational AI. DST plays a crucial role in task-oriented dialogue systems, where it focuses on tracking user goals and slot-value updates throughout the conversation to ensure accurate and contextually appropriate responses. Numerous advancements have been made in dynamic state tracking, allowing dialogue systems to handle evolving user intents and goal shifts effectively.

However, while DST effectively manages explicit goal changes, it does not adequately capture the logical transitions underlying users’ reasoning processes. In real-world conversations, users frequently adjust logic between sentences: introducing assumptions, modifying conclusions, and refining their reasoning based on new information. For example, a user might initially state, "I prefer cheap restaurants" but later refine their reasoning with: "However, if the food quality is poor, I wouldn’t go." Traditional DST would only update the “price

range” = “cheap state”, failing to capture the logical shift where food quality becomes a deciding factor. Existing research on logic tracking has primarily focused on maintaining coherence and consistency within a given context, without explicitly modeling how logic itself evolves over time.

As the study of Tracking Dynamic Logic Transitions in Conversations remains underexplored, our work aims to bridge this gap by explicitly tracking logical shifts, allowing models to capture both factual goal updates and reasoning evolution. This approach would enhance conversational AI by improving logical consistency, adaptability, and response quality.

2 Dataset construction

We used different existing datasets to construct our dataset.

2.1 TARCED

Initially, we noticed that the TARCED dataset is composed of numerous factual statements, each embedding certain logical connections. This observation sparked the idea that these facts could serve as the basis for a question-answer dialogue system. To explore this, we downloaded a pre-annotated training dataset from GitHub. We then filter the entries with the most useful relational labels, such as ‘per:title’, ‘per:origin’, and ‘org:top members/employees’. Next, we reassemble the previously tokenized fragments within these entries into complete sentences.

Using GPT-4, we rewrote these reconstructed sentences into dialogues about the fact in the sentences, ultimately creating two types of dialogues for each entry: one demonstrating a successful logical transformation and one depicting an unsuccessful logical conversion. This process allowed us to systematically investigate how intrinsic logical relationships in factual assertions can be leveraged in a dialogue setting.

2.2 DialogStudio

We focused on preparing DialogStudio for evaluating logical consistency in multi-turn dialogues. To begin, we accessed the MULTIWOZ2.2 subset via the Hugging Face Datasets API, which offers a structured collection of task-oriented conversations annotated across multiple domains such as restaurant booking, train travel, and taxi service. After downloading the dataset, we filtered out lower-quality entries and selected dialogues containing at least four turns and clear user goals to ensure a meaningful context for logic manipulation.

Each dialogue was parsed and reformatted into a standardized JSON structure with alternating "user" and "system" roles. This preprocessing step allowed us to isolate turn indices for logical revision and ensure uniformity across inputs to both the rewriting and evaluation pipelines. We wrote custom scripts to traverse and clean the raw log fields from DialogStudio, extract relevant utterances, and store them in a structured format suitable for downstream logic modification.

2.3 SNLI

The datapoints in SNLI are in the format of two sentences along with an indicator showing the relationship between them (entailment, contradiction, or neutrality). Initially, we planned to use randomly selected sentences as a cluster to generate a dialogue through the ChatGPT API. However, these sentences were either very similar or had no meaningful connection—for example, “The old man is drinking orange juice” vs. “The old man is drinking milk.” This is due to the nature of the dataset: it contains both similar and unrelated sentence pairs to help models learn from small modifications and various semantic cases. As a result, clusters of such sentences led to dialogues with no clear or deep logical chain.

As an alternative, we used single sentences as topics or scenarios and utilized the ChatGPT plugin to generate AI-human dialogues based on these topics. To ensure a coherent logical flow, we added prompts like “The next user query must use information from the last AI response.” In other words, prompt tuning became the key component in this data generation step. Information modifications, final AI responses, and the extraction of logical point lists were all handled by ChatGPT using specific prompts. This generation process is supported by a coding workflow, which makes it easier to scale up

and generate more datapoints in the future.

2.4 Method based on randomly generated word pairs

To construct the original dialogues in this study, we generated word triplets in the form of <subject, relation, object>, within the context of a restaurant scenario. We first asked GPT to independently produce 100 words for each component: subjects, relations, and objects, which are all related to restaurant-specific roles, actions, and items.

These lists were then randomly combined to form 100 unique triplets, for example: { 'subject': 'knife', 'relation': 'clears', 'object': 'milkshake' }, each serving as the semantic anchor for generating a dialogue.

For each triplet, we prompted GPT to create a multi-turn conversation between a user and an AI assistant. To ensure logical consistency and incremental reasoning, we designed the prompting strategy to follow these rules:

- The user initiates the dialogue with a question.
- The AI responds based on the current context.
- Each subsequent user query must build directly on the previous AI response, forming a sequential logic chain.
- After 4–5 turns, the user introduces new information that negates or modifies a previous assumption, and asks the AI to revise its explanation accordingly.

2.5 Logic Modification and Dataset Construction

Currently, original dialogue, final modification user query, and correct revised answer are handled by coding pipeline using Chatgpt plugin. However, the generation of wrong answers is still handled manually through the ChatGPT web interface, since the automated approach struggles to generate convincing incorrect responses. It’s important to note that both the correct and incorrect answers in our dataset are sample answers—meaning it is possible for the AI to generate other valid responses (both correct and true “wrong”). Because of this, We still need to carefully review the wrong answers after each generation. (This is a potential area for future improvement.)

With the combination of this coding workflow and prompt-tuned ChatGPT outputs, we generated

the datapoints and stored them in .jsonl files. Given the length of each datapoint, the .jsonl format allows us to read them one by one, which improves memory efficiency.

This semi-automated rewriting process enables us to efficiently construct a diverse dataset of logic-altered dialogues. The ultimate goal is to use these paired samples to benchmark a range of LLMs on their ability to detect, respond to, and remain consistent with logical shifts in user goals. The rewritten data set forms a key part of our evaluation corpus to test logical consistency in dynamic, goal-driven conversational settings.

3 Dataset Details Report

By this week’s end, we have finished part 1- creating logic consistency datasets. After research and discussion, the final format of datapoints will be composed of:

- A dialogue with multiple exchanges between AI and user. (one exchange includes one user query and one AI response to this query)
- A final user query modifying one piece of information provided during dialogue. This query also asks the AI to repeat and summarize the whole dialogue discussion with the modified information.
- A sample correct final answer of AI for the final user query. (this answer is a paragraph, not a modified dialogue)
- A sample wrong final answer of AI for the final user query.
- Logic shift:
 - A list of key points refers to the logic chain in the original dialogue
 - A list of key points refers to the logic chain in AI’s correct final response after providing information modification.
 - A list of key points refers to the logic chain in AI’s wrong final response after providing information modification.

Noted: this format has changed multiple times when we worked on dialogue generation so it is not an initially finalized one, but a progressively improved version.

Field	Description
dialogue_id	unique identifier of the dialogue instance.
original_context	user-AI multi-turn conversation grounded in a specific scenario.
user_revision	user query that modifies or negates previous information.
correct_revised_response	AI response correctly updated based on user revision.
wrong_revised_response	AI response with fact update but flawed logic.
logical_shift	comparison of logic chains across original, correct, and wrong responses.

Table 1: Structure for each dialogue instance.

See Appendix A for an example JSONL entry illustrating our dialogue structure.

In total, the dataset consists of 62 datapoints encompassing a variety of dialogue lengths, types of logic shifts, and discussion topics. The first two datapoints were designated as reference examples for evaluation purposes, while the remaining 60 datapoints were used for actual evaluation and fine-tuning experiments. Dialogue lengths are categorized based on the number of exchanges (one exchange includes a user query and a corresponding AI response):

- Short: 1–3 exchanges
- Moderate: 4–6 exchanges
- Long: more than 6 exchanges

Table 2 summarizes the distribution of dialogues across these categories.

Dialogue Length	Value
Short	25
Moderate	23
Long	12

Table 2: Number of dialogues in each length category.

4 Evaluation

To assess the ability of various models (GPT-3.5/4.0, T5, LLaMA2, Claude) to maintain logical consistency across dialogue turns under user-induced logical shifts, we designed a multi-stage evaluation framework as follows:

1. **Response Generation:** Each model was first tasked with generating a revised response based on the provided original dialogue context and a user revision. This simulates a real-world requirement for models to flexibly adapt

to updated information within ongoing conversations.

2. **Logic Point Extraction:** After generation, we extracted key logical propositions (“logic points”) from each model’s revised response. This step aimed to represent the underlying factual or inferential structure embedded in the model’s answer. We used GPT-4.0 to assist in logic point extraction, providing it with a clear instruction and examples.

3. **Evaluation Metric:** GPT-4.0 outputs a numerical score calculated as:

$$\text{Score} = \frac{\text{Number of Logically Adapted Points}}{\text{Total Number of Extracted Points}}$$

The logically aligned points must correctly adapt to the new logical chain adjusted by the user’s revision, without contradictions. Each generated response receives a score between 0 and 1, where 0 indicates complete logical failure and 1 indicates fully correct logical adaptation.

4. **Scoring Mechanism:** The scoring was performed automatically by the GPT-4.0 model. The evaluation prompt was structured as follows:

- We provided the following information:
 - Reformatted original conversation $\Rightarrow$ provides full background context.
 - User’s revision $\Rightarrow$ describes the change of previous info.
 - Model-generated response $\Rightarrow$ the output to be evaluated.
 - Extracted logic points from the model’s generated response.
 - Original dialogue logic points $\Rightarrow$ the original logical chain.
 - Sample correct revised response & corresponding logic points $\Rightarrow$ serves as a positive reference.
 - Sample wrong revised response & corresponding logic points $\Rightarrow$ serves as a negative reference.
- explaining the evaluation objective, the scoring criteria, and common types of errors (e.g., failure to update, introducing contradictions).
- providing a full worked example: a sample datapoint with a detailed explanation

of why each logic point is correct or incorrect, as an extra reference for the evaluation.

- asking for a numerical score output, along with a short explanation of this score. This explanation will be used to interpret whether the evaluation is working as expected.

After generating a score and explanation using the evaluation prompt, we applied an additional score-extraction prompt to obtain the numerical score. The extracted score was then recorded for analysis.

5. **Final Metric Computation:** For each model, we computed the average score across all datapoints, representing the model’s overall ability to logically adapt to user revisions throughout the whole dataset.

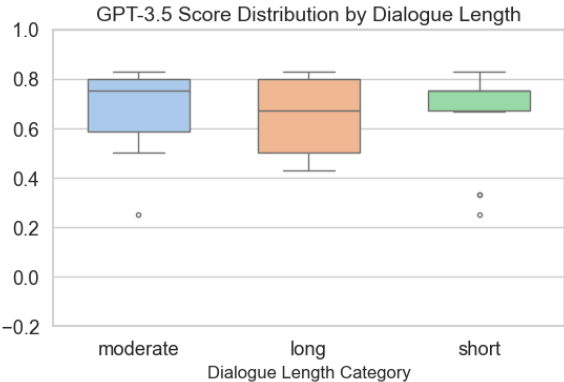
5 Evaluation Result and Analysis

Based on the results in Table 3 (located in next page) and Table 4, Claude achieved the highest average score (0.84), followed by GPT-4 (0.76) and GPT-3.5 (0.67), while T5 had the lowest average score (0.25).

Model	Average Score
T5	0.25
GPT-3.5	0.68
GPT-4	0.76
LLaMA2	0.65
Claude	0.84

Table 4: Average model scores across all dialogues.

5.1 ChatGPT



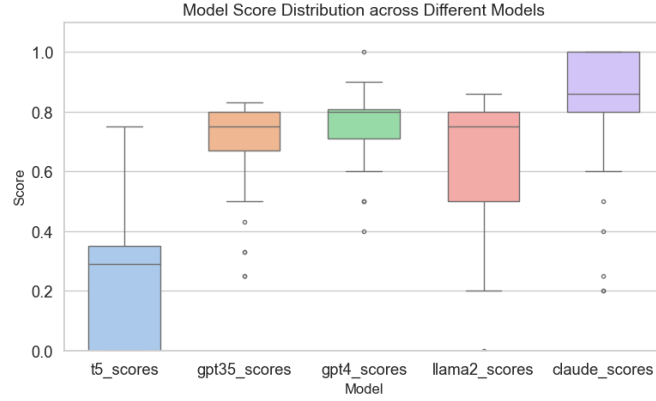
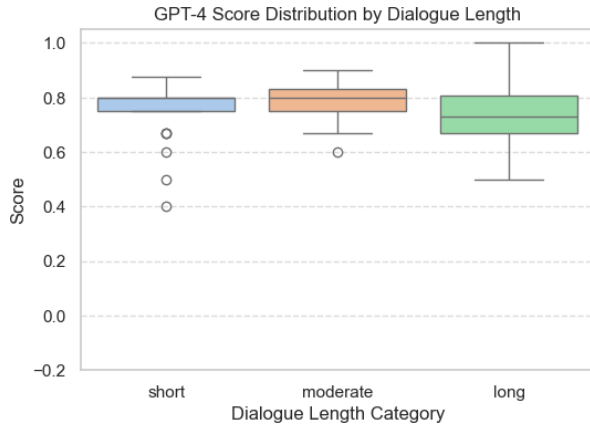


Figure 1: Model Score Distribution across different models.

Dialogue Length	T5	GPT-3.5	GPT-4	LLaMA2	Claude
Short	0.25	0.65	0.75	0.63	0.84
Moderate	0.25	0.69	0.78	0.68	0.86
Long	0.28	0.69	0.73	0.62	0.82

Table 3: Average model scores by dialogue length category.



Overall, both GPT models demonstrate a relatively strong ability to maintain logical consistency after user revisions. The models tend to produce detailed and elaborated answers, often embedding multiple logical propositions ("logic points") within a single response, and the scoring results, ranging from 0.6 to 0.9, indicate that a high proportion of these logic points successfully adapt to the revised context provided by the user. Both GPT-3.5 and GPT-4 demonstrate a relatively strong capability to update and maintain coherent reasoning chains following user-driven logical shifts. This suggests that the GPT family models inherently possess a robust mechanism for incorporating new premises into ongoing dialogues, although minor inconsistencies can still arise, particularly under complex revisions.

In comparison, GPT-4 outperforms GPT-3.5 in terms of stability and robustness. In particular, GPT-4 shows tighter score distributions and fewer low-scoring outliers compared to GPT-3.5. This suggests that GPT-4 is better at consistently adjusting its reasoning logic in response to updated user contexts.

According to the graph, dialogue length does not appear to have a significant impact on performance. Both GPT-3.5 and GPT-4 maintain similar score distributions across short, moderate, and long dialogues. This indicates that the models' ability to track logical consistency is relatively stable regardless of the conversation depth.

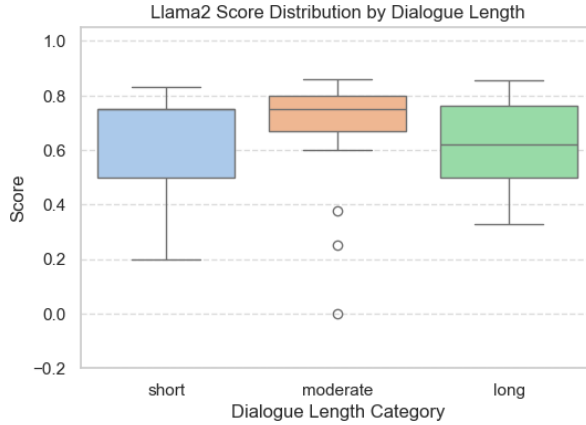
However, GPT-3.5 shows more variance, especially in short dialogues, where some samples drop to much lower scores. This may suggest that GPT-3.5 is more sensitive to short, abrupt logical shifts and less capable of rapid context adaptation compared to GPT-4.

5.2 LLaMA2

The performance of LLaMA2 in our logical consistency evaluation demonstrates performed a relatively high accuracy. After prompting the model with the original dialogue and a revised user query, LLaMA2 was able to generate responses that were grammatically fluent and topically relevant. However, the logical alignment with the original dialogue was often imperfect.

We observed that LLaMA2 responses frequently preserved surface-level coherence but failed to adjust the deeper logic chain based on user modifications. For example, when a user revision contradicted a previous assumption, LLaMA2 occasionally ignored the conflict or offered vague reasoning instead of revising the logic accurately.

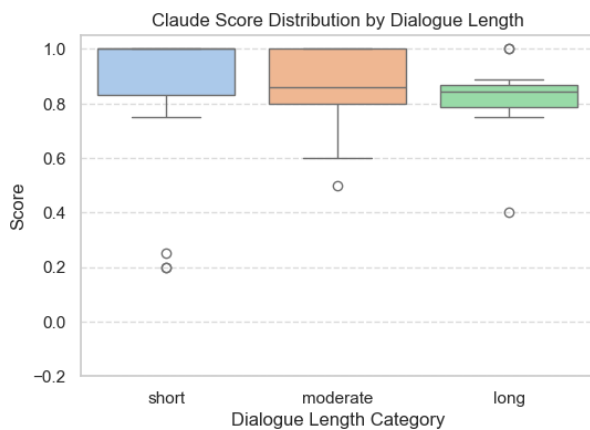
The highest logic scoring results is 0.639, and the score for different dialogue length is as shown below:



This graph shows the dialogue length doesn't play a critical role in generating high logic scoring results.

Nevertheless, the model shows consistency in basic factual continuation and can be a solid foundation for fine-tuning, especially if enhanced with dialogue-specific logic patterns through supervised training.

5.3 Claude



Claude achieved the highest average logical consistency tracking score among all evaluated models, with a mean score of 0.8435 across 60 dialogue instances. The Claude-3-Opus model was prompted to generate revised responses following user-driven logic shifts and evaluated based on its ability to update reasoning chains correctly.

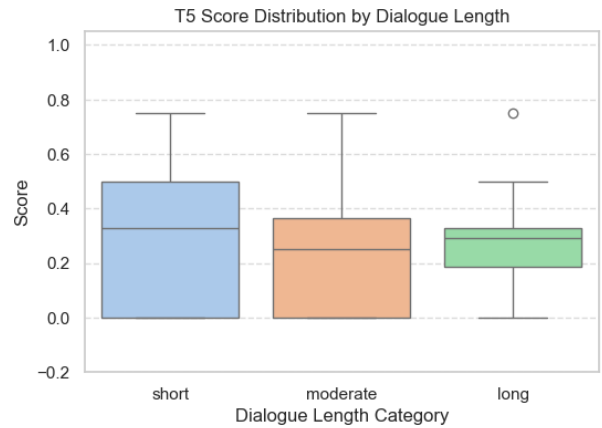
To further analyze Claude's performance, we categorized dialogues by their exchange counts into three groups: short (≤ 3 exchanges), moderate (4–6 exchanges), and long (≥ 7 exchanges). Figure 5 presents the average logical consistency scores across these dialogue length categories.

We found that Claude performed consistently well across all dialogue lengths. Interestingly, moderate-length dialogues (4–6 exchanges) achieved slightly higher average scores compared to short dialogues. Long dialogues also maintained strong performance with only a minimal drop. This suggests that Claude is robust to dialogue complexity and maintains logical coherence even as conversational history becomes longer and requires deeper reasoning across multiple turns.

Qualitatively, Claude responses demonstrated the ability to correctly adjust to user revisions by updating only the relevant portions of logical chains without introducing contradictions or hallucinations. Minor errors, when present, typically involved partial carryovers from outdated user intents rather than major logical breakdowns.

Overall, these results indicate that Claude-3-Opus is highly capable of multi-turn logical consistency tracking, showing promising stability even in more challenging, extended dialogue contexts.

5.4 T5



For the T5 model, we selected FLAN-T5-XL, which is known for its strong general reasoning capabilities. Compared to the other models we evaluated, T5 is not inherently a conversational model, as it does not possess an internal memory mechanism to remember and build upon past interactions. Instead, it is primarily designed for single-exchange tasks. However, we are interested in exploring whether T5 could be adapted for multi-turn dialogue by manually updating the input prompt

with the full conversation history plus the new user query, and whether it could properly adjust its logical chain based on user corrections.

Based on the evaluation results, the selected T5 model demonstrated relatively poor capability in adjusting its logical reasoning chain. In the majority of datapoints, the T5-generated responses merely repeated the original logic without incorporating the user’s new revision or contextual correction. Additionally, the presence of a significant number of zero scores indicates that T5 occasionally failed to update the context, ignoring the modification part within the new user query.

When analyzing the scores across different dialogue lengths, we observed that for longer conversations, the T5 model’s performance deteriorated further. This suggests that as the conversation history becomes longer and more complex, T5 struggles more to maintain the logic consistency and adapt its reasoning to users’ corrections.

6 Fine-Tuning and Analysis

After the evaluation, we used our dataset to fine-tune models and experimented improving logical reasoning within conversational AI models.

6.1 ChatGPT

Since the fine-tuning cost of GPT-4 is relatively high, we chose to fine-tune GPT-3.5 in this study. We first converted our dialogue dataset into the required JSONL format, where the original dialogue context and the user revision were merged to form the user prompt, and the correct revised response served as the assistant’s expected completion.

The fine-tuning process was straightforward: we uploaded the prepared dataset to OpenAI’s platform, initiated fine-tuning on the gpt-3.5-turbo base model, and obtained a newly fine-tuned model.

We then evaluated the fine-tuned model using the same logical consistency scoring method as before. The average score increased from 0.68 to 0.71. Although the improvement is modest, it demonstrates that even with a relatively small dataset, the model was able to learn to better adjust its reasoning chains following user revisions. We anticipate that further improvements could be achieved with larger datasets and extended training.

6.2 LLaMA2

We fine-tuned the LLaMA2 model using instruction tuning on our dataset of logically evolving dia-

logues. The dataset included pairs of original conversations, user revisions, and revised responses. Each training sample was framed as an instruction-prompted generation task, to help the model learn how to adapt system responses based on logical shifts.

To make fine-tuning more GPU efficient, we used Low-Rank Adaptation (LoRA) to update a subset of the model’s parameters and applied 4-bit quantization, making the process possible on using one GPU. The prompts were structured with explicit sections for “Original Dialogue,” “Logic Shift,” and a final instruction asking the model to generate a revised system response.

After fine-tuning, we reran the generation process and extracted logic chains from the outputs. The average logic match score improved compared to the zero-shot setting, which demonstrates that the model had learned to track and update its reasoning in response to user revisions more accurately. The final score for fine-tuned LLaMA2 was 0.71, comparing to the model before, the improvement was not significant, yet still shows a promising progress toward consistent logic tracking in dialogue settings.

6.3 Claude

Due to Anthropic’s current model access policies, external fine-tuning of Claude models is not permitted. Thus, we only evaluated Claude in a zero-shot setting without fine-tuning or adaptation to our task-specific data.

6.4 T5

LR	Batch Size	Grad Accum	Tuned Avg Score
2e-5	2	2	0.276
1e-5	2	4	0.275
1e-5	2	2	0.242
2e-5	2	1	0.247
2e-5	3	2	0.120
2e-5	4	1	0.240
2e-5	5	1	0.257

Table 5: Summary of T5 fine-tuning experiments with different hyperparameters.

Note: Original Score was fixed to 0.267.

To fine-tune, we randomly split the entire dataset (60 datapoints) into a training set (40 datapoints) and a test set (20 datapoints), maintaining the same distribution of dialogue lengths across the three categories (short, moderate, and long). This ensures that the model has a more representative

and balanced training set. We conducted training by setting a slightly modified response generation prompt as the input and using the template `correct_revised_answer` as the target label.

In the fine-tuning experiments on T5, we primarily adjusted three hyperparameters: learning rate, batch size, and gradient accumulation steps. A smaller learning rate generally results in more stable but slower updates. Batch size determines how many datapoints are processed together before updating the model parameters; a larger batch size leads to more stable updates but requires more memory. Gradient accumulation steps allow the model to accumulate gradients over multiple smaller batches to simulate a larger effective batch size. Together, the batch size multiplied by the gradient accumulation steps determines the effective batch size, which controls the overall stability and smoothness of the updates during training.

Across the experiments, the best score obtained was 0.276, representing a slight improvement compared to the original score of 0.267. We also observed that increasing the effective batch size tended to cause worse response generation quality. In short, fine-tuning the T5 model showed very limited improvement in its logical adjustment capability.

7 Discussion and future plan

7.1 Fine Tuning Method for T5

One potential reason for such limited improvement lies in the training process of fine tuning. During fine-tuning, the default loss function (cross-entropy) measures how closely the generated response matches the target response at the token level—that is, based on surface-level similarity rather than deeper logical structure. As a result, the model can learn to better mimic the wording or phrasing of the template correct answer, but does not necessarily understand or adjust the underlying logic. A simple example highlights this issue: inserting a single word like “*not*” into a sentence can completely invert its logical meaning; however, token-level loss would still be very small. Thus, the fine-tuned T5 model’s logical reasoning ability remains largely unchanged because the training process does not explicitly penalize logical inconsistencies. Other fine tuned models may experience a similar but less severe impact from this cause, given their improvement.

In future work, we propose to develop a new

logical-level loss function to better fine-tune the model’s reasoning ability. Instead of relying solely on token-level similarity between generated and target responses, we plan to incorporate logical inconsistency loss. This approach could potentially bridge the gap between surface-level similarity and deeper logical understanding during fine-tuning.

7.2 Size of Dataset

One limitation of our current study is the relatively small size of the fine-tuning dataset. With only 60 dialogue instances, the model’s exposure to diverse logical shift patterns was restricted. As a result, although fine-tuning produced measurable improvements in logical consistency tracking, the overall gains remained modest.

A larger dataset would likely benefit the model in several ways: (1) increasing the diversity of logical revisions the model encounters, (2) reducing overfitting to a narrow set of examples, and (3) enabling better generalization to more complex and subtle user modifications. Our future work will also focus on expanding the dataset both in quantity and in logical complexity to fully leverage the potential of fine-tuning for dynamic reasoning adaptation.

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<https://arxiv.org/abs/2502.07374>.

A Example JSONL Datapoint

```
{
  "dialogue_id": 0,
  "original_context": [
    {
      "speaker": "User",
      "text": "I_am_a_farmer_and_want..."
    },
    {
      "speaker": "AI",
      "text": "In_a_wet_and_humid_climate..."
    },
    {
      "speaker": "User",
      "text": "I_haven't_checked.How..."
    },
    {
      "speaker": "AI",
      "text": "You_can_test_soil_drainage..."
    },
    {
      "speaker": "User",
      "text": "Yes!_What_varieties_are..."
    },
    {
      "speaker": "AI",
      "text": "In_humid_environments,_you..."
    },
    {
      "speaker": "User",
      "text": "Yes!_Since_my_farm_is..."
    },
    {
      "speaker": "AI",
      "text": "In_humid_climates,_less..."
    },
    {
      "speaker": "User",
      "text": "I_haven't_used_mulch_before..."
    },
    {
      "speaker": "AI",
      "text": "Use_light,_well-aerated_mulch..."
    }
  ],
  "user_revision": {
    "speaker": "User",
    "text": "Wait!_I_made_a_mistake-my_farm..."
  },
  "correct_revised_response": {
    "speaker": "AI",
    "text": "1._Managing_Water_Scarcity_->_Ensuring_Soil_Holds_Moisture..."
  },
  "wrong_revised_response": {
    "speaker": "AI",
    "text": "Thanks_for_the_clarification!_In_hot_and_dry_climates..."
  },
  "logical_shift": {
    "original_logic": [
      "Climate=_humid",
      "Focus_on_drainage...",
      "Use_fungal-resistant..."
    ],
    "correct_revised_logic": [
      "Climate=_hot_and_dry",
      "Focus_on_water_retention...",
      "Use_heat-tolerant..."
    ],
    "wrong_revised_logic": [
      "Suggests_drainage_is_a_priority...",
      "Recommends_fungal-resistant...",
      "Advises_daily_watering..."
    ]
  }
}
```

Listing 1: Truncated JSONL entry showing structure